



Original Article

Impact of atrial fibrillation type (paroxysmal vs. non-paroxysmal) on long-term clinical outcomes: The RAFFINE registry subanalysis

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ABSTRACT

Background: The type of atrial fibrillation (AF) (paroxysmal or non-paroxysmal) is important in determining its therapeutic management. However, the prognostic impact of AF type on the incidence of cardiovascular events remains uncertain.

Methods: We investigated patients with AF who were selected from an observational, multicenter, prospective registry (RAFFINE) comprising 4 university hospitals and 50 general hospitals/clinics in Japan between 2013 and 2015. In this subanalysis study, patients were divided into two groups according to their AF pattern at the time of enrollment. The primary outcome was the composite of death, ischemic stroke, and heart-failure-related hospitalization.

Results: Among 3845 patients, 1472 (38.3%) and 2373 (61.7%) had paroxysmal and non-paroxysmal type AF, respectively. Patients with non-paroxysmal AF were older and had higher CHADS₂ score and prevalence of comorbidities. During median follow-up of 3.7 years, 681 (17.7%) primary endpoints were identified. Cumulative incidences of the primary endpoint were significantly higher in the non-paroxysmal AF group; however, rates of bleeding events were not significantly different between the groups. Multivariate Cox hazard analysis showed that non-paroxysmal AF had significantly higher risk of cardiovascular events compared with paroxysmal AF (hazard ratio, 1.38; 95% confidence interval, 1.17–1.64; $p = 0.0002$).

Conclusions: Non-paroxysmal AF was significantly associated with cardiovascular events. Long-term clinical outcomes might be improved if transition from paroxysmal to non-paroxysmal AF can be prevented.

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Introduction

Atrial fibrillation (AF) is the most common type of sustained arrhythmia and its incidence has been increasing worldwide [1]. Although the treatment and management of AF have dramatically improved in the past few decades, AF is associated with an increased incidence of stroke, heart failure, and dementia, as well as risk of mortality [2–6].

The classification of AF into paroxysmal and non-paroxysmal types has gained widespread acceptance in clinical practice, and the type of AF is important in determining the therapeutic management. Paroxysmal pattern is found in more than two thirds of patients newly diagnosed with AF, however, 5 %–10 % of these progress to non-paroxysmal AF per year [3,7]. Thromboembolism is a major risk in patients with AF, and the risk correlates with increasing CHADS₂ or CHA₂DS₂-VASC score, which does not reflect AF type [8,9]. Therefore, it remains unclear whether AF type could be a risk factor of worse clinical events, such as death or thromboembolism, in addition to the traditional risk factors. The aim of the present study was to evaluate the prognostic impact of AF type on clinical outcomes in contemporary practice in Japan, using data from the RAFFINE registry.

Materials and methods

Study subjects

The RAFFINE registry is an observational, multicenter, prospective registry of Japanese patients with AF, which was designed to follow clinical events for at least 3 years and up to 5 years (UMIN Clinical Trials Registry: UMIN000009617) [10]. Five Juntendo University hospitals and 50 general hospitals and clinics (Online Appendix A) participated in the registry. Patient enrollment started in January 2013 and ceased in December 2015. Follow-up was completed in December 2018 [11].

Patients aged 20 years or older with a diagnosis of AF by 12-lead electrocardiogram (ECGs) or 24-hour Holter ECG were eligible for enrollment into the RAFFINE registry. Physicians at each site enrolled consecutive patients with AF who were under regular clinical observation (for at least 3 months) at the outpatient department. Patients with a life expectancy <1 year, hospital in-patients, and those who failed to provide written informed consent were not enrolled. Ethical approval was obtained from the Juntendo University ethics review committee and the relevant ethics committee at each participating site. All patients provided written informed consent. The study was conducted in accordance with ethical principles of the Declaration of Helsinki. Juntendo University institutional review board (IRB) also reviewed the study for institutions that did not have their own internal IRB.

In this registry, the type of AF was classified at baseline by the enrolling physician as paroxysmal (lasting ≤7 days at any time), persistent (duration 1 week to 1 year), or permanent (duration ≥1 year or failed electric cardioversion) AF, as defined by the AF guidelines [1,12]. In the present study, we combined the persistent and permanent subtypes as non-paroxysmal type AF.

Study outcomes

The primary outcome in this study was the composite of all-cause death, ischemic stroke, and heart-failure-related hospitalization. Secondary outcomes included all-cause death, cardiovascular death, ischemic stroke, heart-failure-related hospitalization, and bleeding events (major or clinically relevant non-major bleeding). Clinical outcome events were reported by physicians and verified by document submission, as much as possible. Standardized definitions for selected outcome events are provided in Online Appendix B.

Statistical analysis

We used the database of the RAFFINE registry for this subanalysis. Quantitative data are presented as the mean ± standard deviation (SD) or the median (interquartile range [IQR]). Categorical variables are presented as frequencies. Continuous variables were compared using an unpaired *t*-test or the Mann–Whitney *U* test. Categorical variables (presented as frequencies) were compared using the chi-squared test or Fisher's exact probability test. Unadjusted cumulative

event rates were estimated using Kaplan–Meier curves and compared among groups using log rank tests. Patients were also stratified by age using 75 years as the cut-off and by CHADS₂ score in the Kaplan–Meier analysis. Hazard ratios (HRs) and 95 % confidence intervals (CIs) were determined using Cox proportional hazard models. Univariate and multivariate Cox regression analyses were performed to identify determinants of clinical outcomes. We selected some clinically significant variables and performed univariate Cox hazard analysis of the primary outcomes and variables that had a *p*-value < 0.05 in the univariate Cox hazard analysis were included in the multivariate analysis. The multivariate model was adjusted for age, sex, congestive heart failure, hypertension, diabetes mellitus, history of stroke or transient ischemic attack (TIA), coronary artery disease, peripheral artery disease, creatinine clearance, and direct oral anticoagulant use. Variables showing values of *p* < 0.05 were considered statistically significant. All statistical analyses were performed using JMP version 14.2.0 software (SAS Institute, Cary, NC, USA).

Results

Baseline characteristics

Of the 3889 participants in the RAFFINE registry, 1472 (38.3 %) were classified as paroxysmal and 2373 (62.7 %) as non-paroxysmal (persistent or permanent) at baseline. The baseline clinical characteristics of patients are shown in Table 1. Patients with non-paroxysmal AF were significantly older and more likely to be male, and have congestive heart failure, diabetes mellitus, and renal deficiency compared with those with paroxysmal AF. Past histories of ischemic stroke or TIA, and coronary artery disease were more common in patients with non-paroxysmal AF. In addition, patients with non-paroxysmal AF had higher body mass index, CHADS₂ score, and CHA₂DS₂-VASC score. Blood test data revealed that patients with paroxysmal AF had lower baseline hemoglobin and serum creatinine, as well as higher platelet count levels, compared with those with non-paroxysmal AF. Brain natriuretic peptide level was significantly higher in the non-paroxysmal group than the paroxysmal group (median 137.0 vs. 66.5 pg/ml, *p* < 0.0001). Regarding medication, oral anticoagulants, diuretics, and rate control drugs were used more frequently in the non-paroxysmal AF group. In contrast, patients with paroxysmal AF were more likely to use antiplatelet drugs or statins.

Clinical outcomes

The median duration of follow-up was 3.7 years (IQR: 3.0, 5.0 years). The primary outcome was identified during follow-up in 681 of the overall patients (17.7 %). The results of the cumulative incidence of clinical outcomes are shown in Table 2. Kaplan–Meier analysis showed that the occurrence of the primary outcome was clearly higher in the non-paroxysmal AF group compared with the paroxysmal AF group (20.5 % vs. 13.2 %, log-rank *p* < 0.0001, Fig. 1). Furthermore, Kaplan–Meier curves showed significant differences in the incidence of events between the two AF groups, even when stratified by age cut-off of 75 years and CHADS₂ score (Fig. 2). Incidences of death, cardiovascular death, ischemic stroke, and heart-failure-related hospitalization were also significantly higher in the non-paroxysmal AF group (Table 2). There was no significant difference in the rate of bleeding events (major or clinically relevant nonmajor bleeding) between the groups.

Multivariate analysis for clinical outcomes

Table 3 shows the results of Cox proportional hazard regression analysis for the primary outcome. According to the univariate analysis, type of AF, congestive heart failure, hypertension, diabetes mellitus, history of stroke or TIA, coronary artery disease, peripheral

Table 1
Baseline clinical characteristics.

	Total (n = 3845)	Paroxysmal (n = 1472)	Non-paroxysmal (n = 2373)	p-Value
Age, years	72.7 ± 9.6	71.7 ± 10.1	73.2 ± 9.2	<0.0001
Age ≥75 years	1691 (44.0)	599 (40.7)	1092 (46.0)	0.001
Male sex	2632 (68.5)	957 (65.0)	1675 (70.6)	0.0003
Height, cm	161.9 ± 9.8	161.8 ± 9.8	161.9 ± 9.8	0.96
Weight, kg	62.6 ± 12.8	61.7 ± 12.4	63.2 ± 12.9	0.0005
Body mass index, kg/m ²	23.8 ± 3.7	23.4 ± 3.5	24.0 ± 3.8	<0.0001
Non-valvular atrial fibrillation	3665 (95.3)	1427 (96.9)	2238 (94.3)	0.0002
Congestive heart failure	911 (23.7)	274 (18.6)	637 (26.8)	<0.0001
Hypertension	2795 (72.7)	1062 (72.2)	1733 (73.0)	0.55
Diabetes mellitus	1167 (30.4)	400 (27.2)	767 (32.3)	0.0007
Dyslipidemia	2022 (52.6)	837 (56.9)	1185 (49.9)	<0.0001
Current smoker	342 (8.9)	116 (7.9)	226 (9.5)	0.08
Ischemic stroke or transient ischemic attack	776 (20.2)	267 (18.1)	509 (21.5)	0.01
Coronary artery disease	528 (13.7)	248 (16.9)	280 (11.8)	<0.0001
Peripheral artery disease	126 (3.3)	45 (3.1)	81 (3.4)	0.55
Creatinine clearance, mL/min	65.2 [49.0, 84.1]	66.9 [50.1, 85.7]	64.4 [48.3, 83.3]	0.004
Creatinine clearance <50	1016 (26.5)	360 (24.6)	656 (27.7)	0.03
Hemodialysis	36 (0.9)	16 (1.1)	20 (0.8)	0.44
Pacemaker implantation	278 (7.2)	142 (9.7)	136 (5.7)	<0.0001
Catheter ablation for atrial fibrillation	127 (3.3)	100 (6.8)	27 (1.1)	<0.0001
CHADS ₂ score	2.0 ± 1.3	1.8 ± 1.2	2.1 ± 1.3	<0.0001
CHA ₂ DS ₂ -VAsC score	3.2 ± 1.6	3.0 ± 1.6	3.3 ± 1.6	<0.0001
Blood tests				
Hemoglobin, g/dL	13.6 ± 1.7	13.4 ± 1.6	13.7 ± 1.8	<0.0001
Platelet count, x1000/μL	187 [157, 221]	196 [169, 231]	180 [151, 214]	<0.0001
Aspartate aminotransferase, IU/L	23 [19, 28]	22 [18, 27]	23 [19, 28]	<0.0001
Alanine aminotransferase, IU/L	18 [13, 24]	18 [13, 24]	18 [13, 24]	0.67
Blood urea nitrogen, mg/dL	18.0 ± 7.4	17.5 ± 6.6	18.3 ± 7.8	0.002
Serum creatinine, mg/dL	0.84 [0.70, 1.01]	0.81 [0.68, 0.99]	0.86 [0.72, 1.02]	0.11
Urinary acid, mg/dL	5.9 ± 1.5	5.7 ± 1.5	6.0 ± 1.5	<0.0001
Low-density lipoprotein-cholesterol, mg/dL	101.7 ± 26.7	103.6 ± 26.7	100.5 ± 26.7	0.0007
High-density lipoprotein-cholesterol, mg/dL	54.6 ± 15.2	55.0 ± 14.8	54.5 ± 15.4	0.32
Triglycerides, mg/dL	109 [78, 157]	112 [81, 161]	108 [76, 154]	0.003
Hemoglobin A1c, %	6.0 ± 0.8	5.9 ± 0.7	6.1 ± 0.8	<0.0001
Brain natriuretic peptide, pg/mL	111.0 [58.6, 198.3]	66.5 [37.9, 137.6]	137.0 [82.7, 222.9]	<0.0001
High-sensitivity C-reactive protein, mg/dL	0.09 [0.04, 0.20]	0.08 [0.04, 0.19]	0.09 [0.05, 0.22]	0.25
Baseline drugs				
Oral anticoagulants	3379 (87.9)	1161 (78.9)	2218 (93.5)	<0.0001
Warfarin	1717 (44.7)	474 (32.2)	1243 (52.4)	<0.0001
Direct oral anticoagulant	1662 (43.2)	687 (46.7)	975 (41.1)	0.0007
Antiplatelet drugs	990 (25.8)	409 (27.8)	581 (24.5)	0.02
Diuretics	1030 (26.8)	278 (18.9)	752 (31.7)	<0.0001
ACE-I/ARBs	2038 (53.0)	790 (53.7)	1248 (52.6)	0.52
Rate control drugs	2268 (59.0)	769 (52.4)	1499 (63.2)	<0.0001
Beta-blockers	1047 (27.2)	432 (29.4)	615 (25.9)	0.02
Verapamil and/or Diltiazem	507 (13.2)	189 (12.8)	318 (13.4)	0.62
Digitalis	771 (20.1)	128 (8.7)	643 (27.1)	<0.0001
Rhythm control drugs	902 (23.5)	740 (50.3)	162 (6.8)	<0.0001
Statins	1387 (36.1)	596 (40.5)	791 (33.3)	<0.0001
Antacid agents	1993 (51.8)	769 (52.2)	1224 (51.6)	0.69

ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker.

artery disease, creatinine clearance, and direct oral anticoagulant use were included in the multivariate model. Multivariate Cox hazard analysis showed that non-paroxysmal AF had significantly higher risk of the primary outcome compared with paroxysmal AF (HR, 1.38; 95%CI, 1.17–1.64; $p = 0.0002$). Higher age, male sex, congestive heart failure, diabetes mellitus, history of stroke or TIA, and higher creatinine

clearance level were also significantly associated with incidence of the primary outcome among AF patients.

Table 4 shows the results of multivariate Cox hazard analysis of AF type for clinical outcomes. Non-paroxysmal AF had significantly higher risk of all-cause death (HR, 1.35; 95%CI, 1.08–1.70; $p = 0.009$) and heart failure (HR, 1.48; 95%CI, 1.15–1.92; $p = 0.003$) as well as the primary

Table 2
Cumulative incidence of clinical events.

	Total (n = 3845)	Paroxysmal AF (n = 1472)	Non-paroxysmal AF (n = 2373)	p-Value
Primary outcome, n (%)	681 (17.7)	194 (13.2)	487 (20.5)	<0.0001
Death, n (%)	382 (9.9)	109 (7.4)	273 (11.5)	<0.0001
Cardiovascular death, n (%)	201 (5.2)	57 (3.8)	144 (6.1)	0.003
Ischemic stroke, n (%)	128 (3.3)	35 (2.4)	93 (3.9)	0.01
Heart-failure-related hospitalization, n (%)	310 (8.1)	82 (5.6)	228 (9.6)	<0.0001
Major or clinically relevant nonmajor bleeding, n (%)	286 (7.4)	96 (6.5)	190 (8.0)	0.09

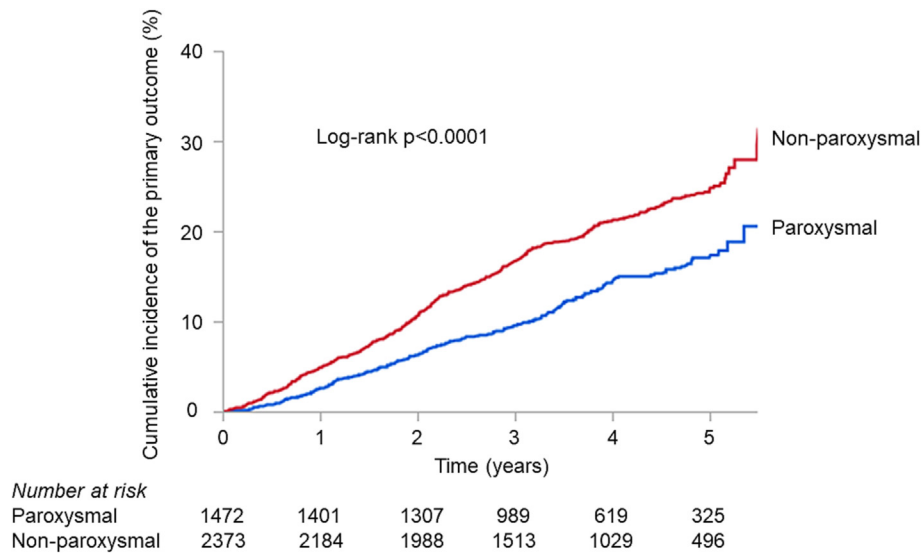


Fig. 1. Kaplan–Meier curves for primary outcome in the overall patients. The occurrence of the primary outcome was clearly higher in the non-paroxysmal AF group compared with the paroxysmal AF group (20.5 % vs. 13.2 %; log-rank, $p < 0.0001$).

AF, atrial fibrillation.

outcome. AF type was not identified as an independent predictor of cardiovascular death, ischemic stroke, or bleeding events in the multivariate Cox hazard analysis.

Discussion

The present study examined the prognostic impact of AF type on clinical outcomes. The major findings were as follows: (1) clinical risk profile was worse in terms of age and underlying comorbidities in the non-paroxysmal AF group compared with the paroxysmal AF group; (2) incidence of the primary outcome was significantly higher in the non-paroxysmal AF group compared with the paroxysmal AF group, and these outcomes were consistent even when stratified by age or

CHADS₂ score; (3) even when adjusted for clinical risk factors, AF type was associated with incidence of the primary outcome, all-cause death, and heart failure.

Previous studies have evaluated the impact of AF type on the incidence of stroke and thromboembolism; however, the results have been inconsistent. Current guidelines on AF management [1] recommend that AF type should not influence the prescription of oral anticoagulants. The report of the Stockholm Cohort of Atrial Fibrillation, which included 1981 AF patients from one of Scandinavia's largest hospitals, showed that the incidence of stroke was similar in paroxysmal and permanent AF (26 vs. 29 events/1000 patient years) during follow-up of 3.6 years [13]. In the EORP-AF General Pilot Registry, the rate of stroke or TIA and thromboembolism was low at 1 year and did not differ between

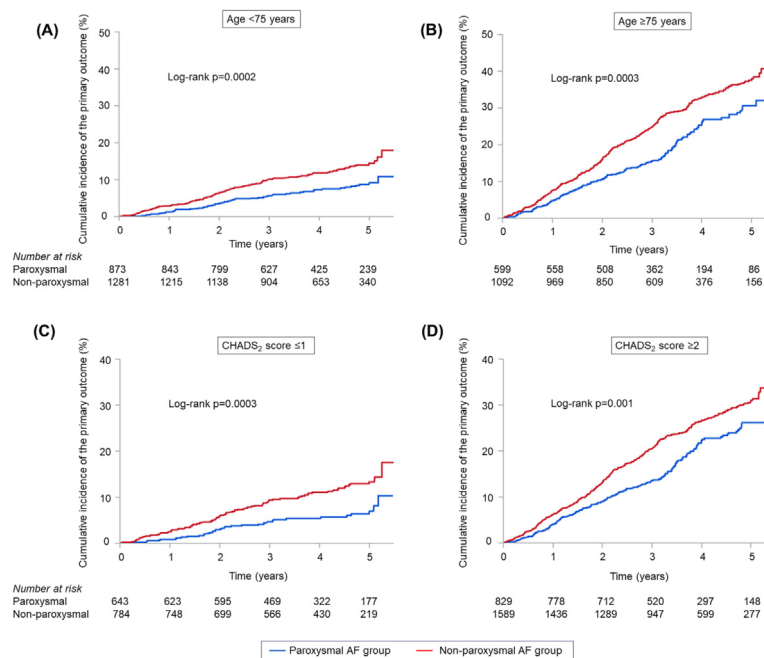


Fig. 2. Kaplan–Meier curves for primary outcome among patients aged <75 years (A), aged ≥75 years (B), with CHADS₂ score ≤1 (C), and with CHADS₂ score ≥2 (D). Significant differences are seen in the incidence of events between the two AF groups, even when stratified with cut-offs for age of 75 years and CHADS₂ score (log-rank, all $p < 0.01$).

AF, atrial fibrillation.

Table 3
Cox hazard analysis for the primary outcome.

	Univariate			Multivariate		
	HR	95 % CI	p-Value	HR	95 % CI	p-Value
Non-paroxysmal (vs. paroxysmal)	1.61	1.36–1.90	<0.0001	1.38	1.17–1.64	0.0002
Age	1.09	1.08–1.10	<0.0001	1.06	1.04–1.07	<0.0001
Male sex	0.85	0.72–0.99	0.04	1.18	1.00–1.40	0.048
Congestive heart failure	2.13	1.83–2.49	<0.0001	1.65	1.40–1.95	<0.0001
Hypertension	1.30	1.08–1.55	0.005	0.998	0.83–1.20	0.98
Diabetes mellitus	1.46	1.25–1.70	<0.0001	1.33	1.13–1.56	0.0005
Dyslipidemia	0.86	0.74–1.00	0.06			
Current smoker	0.85	0.64–1.13	0.27			
Ischemic stroke or TIA	1.56	1.31–1.85	<0.0001	1.28	1.08–1.52	0.005
Coronary artery disease	1.67	1.38–2.01	<0.0001	1.12	0.92–1.37	0.26
Peripheral artery disease	1.81	1.30–2.52	0.0004	1.32	0.94–1.85	0.11
Creatinine clearance	0.97	0.97–0.98	<0.0001	0.99	0.98–0.99	<0.0001
Direct oral anticoagulant use	0.78	0.66–0.91	0.002	0.99	0.84–1.16	0.87

95%CI, 95 % confidence interval; HR, hazard ratio; TIA, transient ischemic attack.

paroxysmal and non-paroxysmal AF [14]. Conversely, results from the ARISTOTLE trial [15], ROCKET-AF trial [16], and ENGAGE AF-TIMI 48 trial [17] showed that patients with non-paroxysmal AF had higher risk of stroke and thromboembolism compared with those with paroxysmal AF. In the present study, the rate of ischemic stroke was significantly higher in patients with non-paroxysmal AF; however, there was less difference between the two groups after adjusting for clinical factors. Patients with paroxysmal AF had lower CHADS₂ score and were less likely to use anticoagulant drugs. If they were taking anticoagulant therapies more frequently, the risk of stroke might be further reduced.

Although patients with non-paroxysmal AF had a worse clinical risk profile, including higher age and higher prevalence of comorbidities, the adjusted hazard risk of the primary outcome was significantly higher in the non-paroxysmal than in the paroxysmal AF group. An et al. demonstrated that non-paroxysmal AF was associated with a higher incidence of cardiovascular events that was seen only in younger patients with AF (≤ 74 years) but not in the older age groups (75–84, and ≥ 85 years) [18]. In the present study, even when stratified by age or CHADS₂ score, differences in the risk of clinical outcomes between groups were consistent. Although the mechanisms by which the type of AF influences cardiovascular outcomes have not been investigated, it has been reported that AF progression is associated with hemodynamic instability and progressive endocardial damage/dysfunction in the left atrium [3]. Longer continuation of AF might cause cardiac dysfunction by higher heart rates or a loss of atrial contraction. In the present study, heart failure events occurred more frequently in the non-paroxysmal AF group. Furthermore, there were significant differences in type of oral anticoagulants between groups. Patients in the paroxysmal AF group were more likely to take direct oral anticoagulants and less likely to use warfarin. Previous reports have shown that direct oral anticoagulants are associated with better clinical outcomes relatively to warfarin [19,20], therefore, direct oral anticoagulants were recommended over warfarin in recent guidelines [21]. Although type of anticoagulants was entered into the multivariate model, these differences between groups might affect clinical outcomes in the present study.

Table 4
Multivariate Cox hazard analysis of non-paroxysmal atrial fibrillation for clinical outcomes.

	HR	95%CI	p-Value
Primary outcome	1.38	1.17–1.64	0.0002
Death	1.35	1.08–1.70	0.009
Cardiovascular death	1.33	0.97–1.81	0.08
Heart-failure-related hospitalization	1.48	1.15–1.92	0.003
Ischemic stroke	1.40	0.94–2.07	0.10
Major or clinically relevant nonmajor bleeding	1.13	0.88–1.46	0.32

95%CI, 95 % confidence interval; HR, hazard ratio.

It is known that paroxysmal AF progresses to non-paroxysmal AF in many patients over time, and the rate of AF progression has been reported as 5 %–10 % per year [3,22,23]. The present study demonstrated the prognostic impact of AF pattern on cardiovascular events; i.e. the findings suggested that stopping AF progression improved patients' clinical outcomes. Rhythm control strategies have been shown to stop or slow progression of AF and might provide an opportunity to improve clinical outcomes. A recent randomized trial compared early rhythm control with usual care in patients who had early AF (defined as AF diagnosed ≤ 12 months before enrollment) and other cardiovascular disease or stroke risk factors [24]. A strategy of initiating rhythm-control therapy was associated with a lower risk of death from cardiovascular causes, stroke, and hospitalization for heart failure or acute coronary syndrome than usual care over a follow-up period of >5 years. To avoid AF progression, rhythm control strategies such as catheter ablation could be considered in patients with paroxysmal AF, particularly in those who have some risk factors for cardiovascular disease.

This study had several limitations. First, as a prospective observational study, unknown confounding factors or selection bias might have affected the outcomes, regardless of the analytical adjustments. Second, most of the participating sites were in the Kanto area and in an urban district in Japan, causing a selection bias. Third, we collected the data regarding medication and type of AF only at the time of enrollment, and some patients might have changed medications or AF type during the follow-up period. Finally, we did not have data of catheter ablation during follow-up and could not adjust effect of this treatment in Cox hazard analysis. Catheter ablation decreases progression of AF and might affect clinical outcomes in AF patients.

Conclusions

Patients with non-paroxysmal type AF had worse clinical risk profiles and clinical outcomes compared with those with paroxysmal type. Even after adjustment for clinical risk factors, AF type became an independent risk factor of cardiovascular events. Long-term clinical outcomes might be improved if the transition from paroxysmal to non-paroxysmal AF can be prevented.

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